7 WASTE MANAGEMENT IMPLICATIONS

7.1 Introduction

7.1.1 This section identifies the types of waste that are likely to be generated during the construction and operation phases of the Project and evaluates the potential waste management implications that may result from waste generated during these phases.

7.1.2 Mitigation measures and good site practices, including waste handling, storage and disposal, have been recommended with reference to relevant waste legislation and management guidelines.

7.1.3 The waste management implications have been assessed in accordance with the requirements outlined in Annex 7 and Annex 15 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) as well as the requirements set out under Clause 3.4.8 of the EIA Study Brief (No. ESB-291/2015).

7.2 Environmental Legislation, Standards and Guidelines

Overview

7.2.1 The relevant legislation, standards and guidelines related to the study for the assessment of waste management implications include:

- Waste Disposal Ordinance (Cap. 354);
- Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C);
- Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N);
- Land (Miscellaneous Provisions) Ordinance (Cap. 28);
- Public Cleansing and Prevention of Nuisances Regulation (Cap. 132BK); and
- Dumping at Sea Ordinance (Cap. 466).

Waste Disposal Ordinance (Cap. 354)

7.2.2 The Waste Disposal Ordinance (WDO) prohibits any unauthorised disposal of waste. Under the WDO, waste can only be disposed of at designated waste disposal facilities licensed by Environmental Protection Department (EPD). The WDO also stipulates the requirements for issuing licenses for the collection and transportation of waste. Breach of this Ordinance can lead to a fine and/or imprisonment.

Waste Disposal (Chemical Waste) (General) Regulation (Cap. 354C)

7.2.3 Chemical waste, as defined under the Waste Disposal (Chemical Waste) (General) Regulation, includes any scrap materials or unwanted substances specified under Schedule 1 of the Regulation, if such a substance or chemical occurs in such a form, quantity or concentration that causes pollution or constitutes a danger to health or risk of pollution to the environment. This Regulation administers the possession, storage, collection, transportation and disposal of chemical waste. EPD has also issued a guidance document, the Code of Practice on the Packaging, Labelling and Storage of Chemical Waste, which details how the chemical waste producers should comply with the regulations on chemical waste.

Waste Disposal (Charges for Disposal of Construction Waste) Regulation (Cap. 354N)

7.2.4 Construction waste as defined under the Waste Disposal (Charges for Disposal of Construction Waste) Regulation includes any substance, matter or thing that is generated from construction work and abandoned, whether or not it has been processed or stockpiled before being abandoned, but does not include any sludge, screening or matter removed in or generated from any desludging, desilting or dredging works. This Regulation stipulated that construction waste delivered to a landfill for disposal must not contain more than 50% by weight of inert material; construction waste delivered to a sorting facility for disposal must contain more than 50% by weight of inert material; whereas construction waste delivered to a public fill reception facility for disposal must consist entirely of inert material.

Land (Miscellaneous Provisions) Ordinance (Cap. 28)

7.2.5 The inert portion of construction and demolition (C&D) materials may be taken to public filling facilities including public filling area, public filling barging points and stockpiling areas. These facilities usually form part of land reclamation schemes and are operated by Civil Engineering and Development Department (CEDD). The Land (Miscellaneous Provisions) Ordinance requires that individuals or companies who deliver public fill to the public filling facilities to obtain Dumping Licences. The licences are issued by CEDD under delegated authority from the Director of Lands.

Public Cleansing and Prevention of Nuisances Regulation (Cap. 132BK)

7.2.6 The Public Cleansing and Prevention of Nuisances Regulation provides control of illegal tipping of waste on unauthorised (unlicensed) sites.

Dumping at Sea Ordinance (Cap. 466)

7.2.7 According to the Dumping at Sea Ordinance (Cap. 466), a permit from EPD is required if any waste producer intend to dump materials from vessels to designated marine dumping areas. The Authority will consider a number of factors including sources and nature of materials to be dumped, dumping rates, need for inspection / testing, water pollution avoidance measures etc. before determining whether such a permit would be granted and, where deemed necessary, any conditions to be complied with. Breach of the requirements in the permit would result in a fine and/or to imprisonment.

Construction & Demolition Materials Management Plan

7.2.8 Measures have been introduced under Section 4.1.3 of the Project Administrative Handbook for Civil Engineering Works (2014 Edition) published by CEDD to enhance the management of C&D materials and to minimise their generation at source. The enhancement measures include (i) drawing up a Construction and Demolition Materials Management Plan (C&DMMP) at an early design stage to minimise C&D materials generation and encourage proper management of such materials; (ii) vetting of the C&DMMP prior to upgrading of the project to Category A in the Public Works Programme; and (iii) providing the Contractor with information from the C&DMMP in order to facilitate the preparation of the Waste Management Plan (WMP) and to minimise C&D materials generation during construction. Projects generating less than 50,000 m^3 C&D materials or importing less than 50,000 m^3 of fill material are exempted from the C&DMMP. The endorsement letter of the C&DMMP of the Project is attached in **Appendix 7.1**.

7.2.9 ETWB TC(W) No. 19/2005 Environmental Management on Construction Sites also includes procedures on waste management requiring the Contractor to reduce the disposed C&D materials. The Contractor is required to prepare and implement an Environmental Management Plan (EMP), which includes the WMP. Further measures have been introduced under DEVB TC(W) No. 6/2010 Trip Ticket System for Disposal of Construction and Demolition Materials, which requires the Contractor to obtain permission from the Public Fill Committee (PFC) in deposing C&D materials to a designated public fill facility.

Other Relevant Guidelines

7.2.10 The following guidelines are also relevant to waste management in Hong Kong:

- WBTC No. 2/93, Public Dumps;
- WBTC No. 2/93B, Public Filling Facilities;
- WBTC Nos. 4/98 and 4/98A, Use of Public Fill in Reclamation and Earth Filling Projects;
- WBTC No. 12/2000, Fill Management;
- WBTC No. 19/2001, Metallic Site Hoardings and Signboards;
- WBTC No. 12/2002, Specification Facilitating the Use of Recycled Aggregates;
- ETWB TC(W) No. 34/2002, Management of Dredged / Excavated Sediment;
- ETWB TC(W) No. 19/2005, Environmental Management on Construction Sites;
- DEVB TC(W) No. 6/2010, Trip Ticket System for Disposal of Construction and Demolition Materials;
- DEVB TC(W) No. 8/2010, Enhanced Specification for Site Cleanliness and Tidiness;
- DEVB TC(W) No. 2/2011, Encouraging the Use of Recycled and other Green Materials in Public Works Projects;
- DEVB TC(W) No. 9/2011, Enhanced Control Measures for Management of Public Fill;
- CEDD TC No. 3/2015, Management of Construction and Demolition Materials;
- Code of Practice on the Packaging, Labelling and Storage of Chemical Waste;
- Code of Practice on the Handling, Transportation and Disposal of Asbestos Waste;
- ProPECC PN 2/97 Handling of Asbestos Containing Materials in Buildings;
- Project Administration Handbook for Civil Engineering Works; and
- Hong Kong Planning Standards and Guidelines, Chapter 9 (Section 6 – Waste Management).

7.3 Description of the Environment

7.3.1 The Project covers an area of about 714 ha and is predominantly rural in character. The existing population at the proposed location of the Project is approximately 42,000. The area has a mixed urban-rural character. Land to the north is predominantly occupied by brownfield operations with some village developments, whereas land in the south is mainly occupied by low-density residential and village developments with some scattered industrial uses.

7.3.2 The existing solid waste arising from the Project area includes domestic waste generated from village houses, commercial and industrial waste generated from open storage and industrial areas and chemical waste generated from vehicle maintenance workshop , open area storage, container storage and recycling facilities. The rural residential area in the middle of the Project area will be retained, while the rest of the Project area will be residential and village neighbourhoods in the north and northeast, "Logistics, Enterprise and Technology Quarter" in the northwest, "Regional Economic and Civic Hub" in the southwest and "District Commercial Node" in the southeast. The contribution of waste generated from these areas is considered in the EIA.

7.3.3 Major existing waste facilities serving the assessment area are shown in **Table 7.1**.

Table 7.1 Summary of Existing Facilities Serving the Assessment Area

Waste Facilities	Location	Date of Commission	Design Capacity
Strategic Landfill			
West New Territories (WENT) Landfill	Nim Wan, Tuen Mun	1993	61 Mm^3
Refuse Transfer Station			
North West New Territories Refuse Transfer Station (NWNT RTS)	Shun Tat Street, Tuen Mun (near Lam Tei)	2001	1,100 tonnes per day
Special Waste Facilities			
Chemical Waste Treatment Centre (CWTC)	51 Tsing Yi Road South, Tsing Yi	1993	100,000 tonnes per year
Sludge Treatment Facility (STF)	Nim Wan, Tuen Mun	2015	2,000 tonnes per day

7.4 Assessment Methodology

7.4.1 The potential waste management implications of the Project have been assessed through:

- Estimation of types and quantities of the waste generated;
- Assessment of potential impacts from the management of the waste with respect to potential hazards, air and odour emissions, noise, wastewater discharge and public transport;
- Examination of the opportunities for reducing waste generation;
- Identification of disposal options for each type of waste; and

- Assessment of impacts on the capacity of waste collection, transfer and disposal facilities.

7.4.2 Prior to considering the disposal options for various types of waste, opportunities for reducing waste generation, on-site or off-site re-use and recycling have been evaluated. Measures which can be taken in the planning and design phases (e.g. by modifying the design approach) and in the construction phase for maximising waste reduction have been separately considered.

7.4.3 After considering all the opportunities for reducing waste generation and maximising re-use, the types and quantities of the waste required to be disposed of have been estimated and the disposal options for each type of waste have been described. The disposal method recommended for each type of waste has taken into account the result of the assessment. The impacts caused by handling (including stockpiling, labelling, packaging and storage), collection, and reuse / disposal of waste have been addressed and appropriate mitigation measures have been proposed.

7.5 Identification and Evaluation of Waste Management Implications

Construction Phase

7.5.1 In accordance with Annex 7 and Annex 15 of the EIAO-TM as well as the requirements set out under Clause 3.4.8 of the EIA Study Brief (No. ESB-291/2015), an application for an Environmental Permit (EP) will be submitted for a number of Schedule 2 Designated Projects (DPs) under this Project. The following activities, including the DPs under Schedule 2 and Schedule 3 and some non-DP elements in the Revised RODP (**Sections 1.5 and 2.4**), have been included in the waste management implication assessment for the construction phase:

- Construction of a new fresh water service reservoir (FWSR);
- Construction of a flood retention lake and flood retention facilities;
- Construction of a new district cooling system (DCS) – subject to further review;
- Construction of a new primary distributor road (Road P1) (DP1);
- Construction of eight new distributor roads (Roads D1 to D8) (DP2);
- Construction of new West Rail Hung Shui Kiu (HSK) Station (Site 4-34) (DP3);
- Construction of Environmentally Friendly Transport Services (EFTS) (DP4) – subject to further review;
- Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and Kong Sham Western Highway (KSWH) connection to Road D3 (DP5);
- Construction of partly depressed and partly decked-over roads located at Road D2, Road D4 and Road D6 (DP6);
- Construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) (DP7) – subject to further review;
- Construction of a new HSK sewage treatment works (STW) (Site 3-26 and part of existing San Wai STW) (DP8);

- Construction of four new sewage pumping stations (SPSs) (Sites 2-34, 3-41, 3-48 and 4-35) (DP9);
- Construction of Flushing Water Service Reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40) (DP10);
- Construction of a new refuse transfer station (RTS) (Site 3-12) (DP11);
- Construction of Road P1 and a slip-road from KSWH to Road D3 partly located within the "Conservation Area" ("CA") of Yuen Tau Shan (DP12);
- "Residential" ("R") development, "Government, Institution or Community" ("G/IC"), "Education" ("E"), "Green Belt" ("GB"), open space, "Amenity" ("A"), other specified uses, industry, roads and "Village Type Development" ("V"); and
- Other miscellaneous construction works, e.g. buildings, roads, utilities, etc.

7.5.2 Typical waste types arising from the proposed works are identified in this section, together with an evaluation of the potential waste management impacts associated with the handling and disposal of waste. **Table 7.2** lists out the sources and examples of the identified waste types.

Table 7.2 Identification of Waste Types during the Construction Phase of the Project

Waste Type	Source of Waste	Example of Waste
Construction and Demolition (C&D) Materials	• Materials generated from site clearance and site formation works • Materials generated from construction of new buildings and infrastructures	• Non-inert C&D materials ➢ Top soil, vegetation and wood waste, etc. ➢ Bamboo, timber, paper and plastic, etc. • Inert C&D materials ➢ Soft materials ➢ Artificial hard materials ➢ All grade granite
Chemical Waste	• Building demolition • Plant operation and maintenance • Maintenance of mechanical equipment	• Asbestos containing materials • Oil and grease, scrap batteries, used paint, fuel, etc. • Cleansing fluids and solvents from construction plant and equipment
General Refuse	• Refuse generated from construction works and site-based staff and workers	• Food waste, containers, cans and waste paper, etc.
Excavated Sediment	• Excavated sediment generated from the inland water removal / diversion works	• Land-based sediment
Contaminated Soil	• Contaminated soil from existing land uses, e.g. vehicle maintenance workshop, open area storage, container storage and recycling facility, etc.	• Contaminated soil

Notes:
1. Non-inert C&D material includes, but not limited to, bamboo, timber, paper and plastic, etc.
2. Soft material includes, but not limited to, top soil, excavated soil, fill, etc.
3. Artificial hard material includes, but not limited to, broken concrete, asphalt, bitumen and granular materials, etc.
4. Granite includes, but not limited to, all grades and types of rock.

Construction and Demolition Materials

7.5.3 The construction phase of the Project will be implemented in a number of development stages through a number of construction activities. The anticipated timing for major construction activities in each development stage is summarised in **Table 7.3**.

Table 7.3 Anticipated Timing for Major Construction Activities in Each Development Stage

Construction Activities	Development Stage	Anticipated Timing
Site Clearance and Site Formation Works	Advance Works	2019 – 2026
	Stage 1	2022 – 2025
	Stage 2	2026 – 2031
	Stage 3	2031 – 2035
	Stage 4	2031 – 2036
Construction of New Buildings and Infrastructures	Advance Works	2021 – 2029
	Stage 1	2022 – 2025
	Stage 2	2026 – 2031
	Stage 3	2031 – 2035
	Stage 4	2031 – 2038

7.5.4 C&D materials will be generated from site clearance and site formation works and construction of new buildings and infrastructure. These C&D materials will comprise both non-inert and inert components.

Site Clearance and Site Formation Works

7.5.5 Site clearance waste will mainly come from the demolition of existing structures, tree felling and preparation of the existing ground surface and will comprise top soil, vegetation, broken concrete and asphalt. Site formation waste will mainly come from the excavation works and will comprise excavated soil and granite. The estimated volumes of C&D materials generated from site formation works are shown in **Table 7.4**.

Table 7.4 Estimated Volumes of C&D Materials Generated from Site Clearance and Site Formation Works

Development Stage	Volume of Non-Inert C&D Material (m^3)	Volume of Inert C&D Material (m^3)			
		Soft Material	Artificial Hard Material	Granite	
				Grade II or above	Grade III or below
Advance Works	22,117	340,150	13,176	13,963	13,105
Stage 1	2,246	155,787	5,242	33,968	29,643
Stage 2	83,738	1,722,995	63,161	173,320	163,741
Stage 3	53,291	1,246,567	44,112	26,756	22,452
Stage 4	20,469	479,579	18,725	16,040	16,040
Total	**181,861**	**3,945,078**	**144,416**	**264,047**	**244,981**

7.5.6 It is estimated that around 181,861 m^3 of non-inert C&D materials, 3,945,078 m^3 of soft materials, 144,416 m^3 of artificial hard materials and 509,028 m^3 of granite (both grade II or above and grade III or below) will be generated from site clearance and site formation works. Some non-inert C&D materials generated from the Project will be suitable for reuse on-site and some will be reused in other projects. Disposal at the WENT Landfill should only be considered as the last resort. Most inert C&D materials generated from the Project are assumed to be suitable for reuse on-site as backfilling materials and only 5,047 m^3 of inert C&D materials will be transported to other concurrent projects for reuse. With proper implementation of good construction site practice and mitigation measures, the on-site handling and reuse of site clearance waste would not cause adverse environmental impacts.

7.5.7 It is estimated that around 1,372,280 m^3 of fill materials will need to be imported during the construction phase of the Project. The reusable portion of the inert C&D materials generated from the site clearance and site formation works will be reused on-site as backfilling materials. The Contractor should review the programme during early construction stage to maximise the quantity of on-site reuse of surplus fill materials. The estimated cut and fill volumes for the development by year are shown in **Table 7.5**.

Table 7.5 Estimated Cut and Fill Volumes for the Development by Year

Year	Cut Volume (m^3) (A)	Fill Volume (m^3) (B)	Net Export / Import of C&D Material (m^3) * (A) - (B)	Import Volume of Fill Material (m^3)	Cumulative Stockpiling Volume (m^3)
2019	826	800	26	0	26
2020	826	4,526	-3,700	3,674	0
2021	3,475	10,907	-7,432	7,432	0
2022	42,736	32,343	10,393	0	10,393
2023	80,177	51,925	28,252	0	38,645
2024	178,189	188,397	-10,208	0	28,437
2025	168,124	201,545	-33,421	4,984	0
2026	736,633	736,211	422	0	422
2027	605,950	554,247	51,703	0	52,125
2028	629,739	554,247	75,492	0	127,617
2029	105,106	20,672	84,434	0	212,051
2030	105,106	20,672	84,434	0	296,485
2031	314,195	612,782	-298,587	2,102	0
2032	540,493	1,131,970	-591,477	591,477	0
2033	650,473	1,244,411	-593,938	593,938	0
2034	318,987	487,660	-168,673	168,673	0
2035	113,734	112,441	1,293	0	1,293**
2036	3,754	0	3,754	0	3,754**
Total	**4,598,523**	**5,965,756**	**-1,367,233**	**1,372,280**	**[** 5,047 to be reused in other concurrent projects]**

Construction of New Buildings and Infrastructure

7.5.8 C&D materials will also be generated from construction of new buildings and infrastructures including the construction of primary and district distributor roads, SPS and associated rising mains, FWSR, reservoirs for reclaimed water use, flood retention facilities and other miscellaneous infrastructure works and will comprise non-inert C&D materials, brick and concrete. The estimated volumes of C&D materials generated from construction of new buildings and infrastructures are shown in **Table 7.6**.

Table 7.6 Estimated Volumes of C&D Materials Generated from Construction of New Buildings and Infrastructures

Development Stage	Gross Floor Area (m^2)	Total C&D Material Generated (m^3)	Volume of Non-Inert C&D Material (m^3)	Volume of Inert C&D Material (m^3)
Advance Works	1,377,300	137,700	27,500	110,200
Stage 1	531,100	53,100	10,600	42,500
Stage 2	4,564,300	456,400	91,300	365,100
Stage 3	4,239,900	424,100	84,800	339,200
Stage 4	1,215,200	121,500	24,300	97,200
Total	**11,927,800**	**1,192,800**	**238,600**	**954,200**

7.5.9 In accordance with the Reduction of Construction Waste Final Report published by the Hong Kong Polytechnic University in March 1993, a C&D materials generation rate of 0.1 m^3 per 1 m^2 of gross floor area (GFA) is adopted. The total estimated GFA of the proposed development in the assessment area is around 11,927,800 m^2. Therefore, it is estimated that around 1,192,800 m^3 of C&D materials would be generated from construction of the buildings and structures for the proposed development.

7.5.10 In addition, based on Monitoring of Solid Waste in Hong Kong (1997) published by EPD, a 2 : 8 non-inert C&D materials: inert C&D materials is adopted. The inert C&D materials is assumed to be all artificial hard materials in the estimation. It is estimated that around 238,600 m^3 of non-inert C&D materials and 954,200 m^3 of artificial hard materials will be generated from construction of new buildings and infrastructure. The non-inert and inert C&D materials generated from construction of new buildings and infrastructure will be reused and recycled within the sites as much as possible before disposal to the WENT Landfill and the Tuen Mun Area 38 Fill Bank, respectively. The generation of C&D materials can be minimised through careful planning during the detailed design stage and with good site practice during construction. This includes the use of non-timber formwork and temporary works and on-site sorting of the C&D materials for reuse and recycling as far as practicable. With proper implementation of good construction site practice and mitigation measures, the on-site handling and reuse of C&D materials would not cause adverse environmental impacts.

Temporary Stockpiling Areas

7.5.11 As shown in **Table 7.5**, the inert C&D materials generated from site clearance and site formation works have to be temporarily stored within the Project site during Advance Works, Stage 1 and Stage 2 of the Project. Since the Project will be implemented in stages, temporary stockpiling areas have been identified to store the C&D materials for reuse under the Project.

7.5.12 The estimated volume of stockpiling for Advance Works Phase 1 in 2019 is about 26 m^3. Since the generation of C&D materials is not significant, it is proposed to store the inert C&D materials temporarily in the works area within the Project.

7.5.13 The estimated volume of stockpiling for Advance Works Phase 2 and Stage 1 from Year 2022 to 2024 is around 40,000 m^3. The size and holding capacity of the temporary stockpiling area located at Site 3-44 are approximately 1.8 hectare and 63,000 m^3, respectively, and are sufficient for storing the inert C&D materials generated from site clearance and site formation works during Advance Works Phase 2 and Stage 1 of the Project.

7.5.14 The estimated volume of stockpiling for Advance Works Phase 3 and Stage 2 from Year 2026 to 2030 is around 300,000 m^3. The size and holding capacity of the temporary stockpiling area located at Sites 2-32, 3-18 and 4-31 are approximately 11.6 hectare and 410,000 m^3, respectively, and are sufficient for storing the inert C&D materials generated from site clearance and site formation works during Advance Works Phase 3 and Stage 2 of the Project.

7.5.15 The storage and stockpiling of C&D materials prior to utilisation on-site may contribute to the generation of dust, visual impacts from unsightliness and water quality impacts from runoff. The disposal of C&D materials also has the potential to result in noise and dust impacts from loading and unloading and emissions from haul vehicles. Mitigation and control requirements for C&D waste are detailed in **Sections 7.6.7 to 7.6.8**. Provided that the handling, storage and disposal of C&D waste are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of C&D waste during the construction phase of the Project are not expected.

Chemical Waste

7.5.16 Asbestos containing materials (ACM) can be found in buildings built before the mid-1980s. If the buildings or structures containing ACM need to be demolished, the ACM should be removed in accordance with the requirements of the Air Pollution Control Ordinance and disposed of in accordance with the requirements of Waste Disposal Ordinance. A Registered Asbestos Consultant and Registered Asbestos Laboratory shall be engaged to conduct investigation for the presence of ACM. An Asbestos Investigation Report, an Asbestos Abatement Plan (if required) and a notification of commencement of asbestos abatement works shall be submitted to EPD at least 28 days before the asbestos abatement works commences. Also, the removal of ACM should be carried out by a Registered Asbestos Contractor according to the approved Asbestos Abatement Plan under the supervision of a Registered Asbestos Consultant. The asbestos waste generated should be disposed of by a licensed chemical waste collector in compliance with the Waste Disposal Ordinance.

7.5.17 The maintenance and servicing of plant, equipment and vehicles will also generate a small amount of chemical waste during the construction phase of the Project. The possible chemical waste includes:

- Scrap batteries from vehicle maintenance;
- Spent hydraulic fluids and waste fuel from plant operation;
- Spent lubrication oils and cleaning fluids from plant maintenance; and
- Spent paint and solvents from equipment maintenance.

7.5.18 Chemical waste arising during the construction phase of the Project may pose environmental, health and safety hazards if not stored and disposed of. The potential environmental, health and safety hazards include:

- Toxic effects to workers;

- Adverse impacts on water quality and aquatic biota from spills; and
- Fire hazard.

7.5.19 It is difficult to quantify the amount of chemical waste that will arise during the construction phase of the Project since it will be highly dependent on the Contractor's on-site construction activities and maintenance practices. Nevertheless, it is anticipated that the quantity of chemical waste, such as lubrication oil and solvent produced from plant and equipment maintenance, would be small and in the order of a few cubic metres per month. The amount of chemical waste to be generated would be quantified in the WMP to be prepared by the Contractor.

7.5.20 Storage, handling, transport and disposal of chemical waste should be arranged in accordance with the Code of Practice on the Packaging, Labelling and Storage of Chemical Waste published by the EPD. Wherever possible opportunities should be taken to reuse and recycle materials. Mitigation and control requirements for chemical waste are detailed in **Sections 7.6.9 to 7.6.14**. Provided that the handling, storage and disposal of chemical waste are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of chemical waste during the construction phase of the Project are not expected.

General Refuse

7.5.21 During the construction phase of the Project, the workforce will generate general refuse comprising food waste, waste paper, empty containers, etc. Storage of general refuse may give rise to adverse environmental impacts, such as windblown litter, odour, water and visual impacts, if not properly managed. The site may also attract vermin and pests if the waste containers are not cleaned or maintained properly and frequently. In addition, disposal of waste at sites other than the approved waste disposal facilities may lead to similar adverse environmental impacts to those sites.

7.5.22 The number of construction workers is not available at this stage but it is anticipated that there will be not more than 3,000 staff to be presented on-site at any one time during the construction phase of the Project. Based on a generation rate of 0.65 kg per worker per day, around 1,950 kg of general refuse would be generated daily during the construction phase of the Project.

7.5.23 In order to minimise the final disposal quantities of general refuse, provision of sufficient number of recycling bins for the collection of different types of recyclable waste (including paper, aluminium cans, plastic bottles and glass bottles) and sufficient number of general refuse bins for the collection of non-recyclable waste is recommended. The Contractor should implement an education programme for workers relating to avoiding, reducing, reusing and recycling of general refuse. A reputable licensed collector should be employed to collect the general refuse on a daily basis for disposal at the NWNT RTS or the WENT Landfill.

7.5.24 Mitigation and control requirements for general refuse are detailed in **Section 7.6.15**. Provided that the handling, storage and disposal of general refuse are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of general refuse during the construction phase of the Project are not expected.

Excavated Sediment

7.5.25 As discussed in **Chapter 5**, a minor portion of the existing inland watercourses will be removed and diverted to the new drainage network to be built under the Project. Also, some existing small ponds and agricultural wet areas will be removed due to the new developments recommended for the Project. Construction works at the existing watercourses / ponds / wet areas within the Project area may result in the small amount of excavated sediment. The excavated sediment to be generated from the construction will consist of soil, small poldered rocks and river / pond sediment.

7.5.26 Information on the quantity and quality of river / pond sediment in the area is not currently available. Metal-based algaecides including copper-based algaecides are not commonly used for fresh water fish farming in Hong Kong due to the high cost and toxicity to fish. Calcium oxide, which is cheaper, non-toxic to fish and a more readily available chemical, is more commonly used to control algae and suspended solids in fish ponds. It is therefore considered unlikely that the excavated sediment will be contaminated. The excavated sediment, however, is expected to have high water content and high organic content. The excavated sediment is proposed to be stabilised / solidified by mixing with cement so that the mixture is suitable for reuse on-site. The handling of this mud may cause odour nuisance and water quality impacts if not properly managed.

7.5.27 All excavated sediment generated from the inland water removal / diversion works should be collected and handled in compliance with the Waste Disposal Ordinance. Soil mixing or cement mixing work is suggested to improve the physical properties of the excavated sediment such that the grading and plasticity of the mixture will be suitable for reuse on-site as backfilling materials in the Project. Mitigation and control requirements for excavated sediment are detailed in **Section 7.6.16**. Provided that the handling of excavated sediment are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling of excavated sediment during the construction phase of the Project are not expected.

Contaminated Soil

7.5.28 Based on site appraisal, 480 potentially contaminated sites were identified during the land contamination assessment (**Chapter 8**). However, the land contamination issues within these concerned sites were considered surmountable due to the following factors:

- Size of individual sites and related operation scale are relatively small and the extent of any potential contamination is anticipated to be localised rather than widespread;
- Chemicals of Concern identified are readily treatable using established physical, chemical and biological techniques; and
- Local remediation experience is sufficient to deal with the nature of the possible contaminants.

7.5.29 Specific site investigation plans, including the proposed soil and groundwater sampling and testing strategy, were presented in the Contamination Assessment Plan (see **Appendix 8.1**). It was found that the major types of land use identified at these potentially contaminated sites at present include vehicle maintenance workshop, open area storage, container storage and recycling facility, etc. Since the development will only commence in phases starting from 2019, there may be change in land use prior to development within both the potentially contaminated and non-contaminated sites. Site re-appraisal should be carried out to determine the latest land uses at the time of land resumption.

7.5.30 If additional potentially contaminated sites are identified, supplementary Contamination Assessment Plan should be prepared and submitted to EPD for approval prior to conducting any site investigation works. Contamination Assessment Report, and if necessary, Remediation Action Plan and Remediation Report, should be prepared subject to the consent from EPD. All submissions and remediation works should be completed before the commencement of the construction works; therefore, no residual impact is anticipated due to presence of contaminated soil within the Project area.

7.5.31 Also, it is proposed to treat contaminated soil by mixing soil with binding agents, e.g. cement such that it can be reused on-site as filling materials. The contaminated soil should be remediated and reused onsite as far as practical; and landfills disposal should be adopted only as a last resort in accordance with Section 4.1 of the Practice Guideline for Investigation and Remediation of Contamination Land. Mitigation and control requirements for contaminated soil are detailed in **Section 7.6.17**. Provided that the handling of contaminated soil are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling of contaminated soil during the construction phase of the Project are not expected.

Transportation Arrangement for Waste Disposal during Construction Phase

7.5.32 Land transport should be used to deliver and dispose of the waste generated from the Project area to the designated disposal outlets. It is expected there will be a maximum of 300 vehicles per day for transporting waste during the construction phase of the Project. The tentative transportation routings for the disposal of various types of wastes generated during the construction phase of the Project are shown in **Table 7.7**. No barging points or conveyor systems will be established in the Project area. The transportation routings may change subject to the actual traffic conditions of the roads. Nevertheless, with the implementation of appropriate mitigation measures (e.g. using water-tight containers and covered trucks), no adverse environmental impacts are expected due to the transportation of waste.

Table 7.7 Tentative Transportation Routings for Waste Disposal During Construction Phase

Disposal Outlet	Type of Waste	Tentative Transportation Routing
Tuen Mun Area 38 Fill Bank	Inert C&D Materials	Via Kong Sham Western Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road and Lung Mun Road
WENT Landfill	Non-inert C&D Materials and General Refuse	Via Kong Sham Western Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road and Nim Wan Road
CWTC	Chemical Waste	Via Kong Sham Western Highway, Yuen Long Highway, Tuen Mun Road, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road

Construction Phase Waste Summary

7.5.33 **Table 7.8** provides a summary of the waste types likely to be generated during the construction phase of the Project, together with the recommended handling and disposal methods.

Table 7.8 Summary of Waste Arising, Waste Handling Procedures and Disposal Routes during the Construction Phase of the Project

Waste Type	Generated from	Materials Generated	Total Amount Generated	Handling Procedures	Disposal Routes
Construction and Demolition (C&D) Materials	• Materials generated from site clearance and site formation works	• Non-inert C&D materials ➢ Top soil, vegetation and wood waste, etc. ➢ Bamboo, timber, paper and plastic, etc.	• 181,861 m^3	• Reusable materials should be separated and recycled as far as practicable	• Reused on-site as much as possible. Materials that cannot be reused nor recycled will be disposed of at the WENT Landfill
		• Inert C&D materials ➢ Soft materials ➢ Artificial hard materials ➢ All grade granite	• 4,598,522 m^3	• Reusable materials should be separated and recycled as far as practicable	• Sorted materials will be stored at the temporary stockpiling areas and reused as much as possible before being disposed of at Tuen Mun Area 38 Fill Bank for beneficial use
	• Materials generated from construction of new buildings and infrastructures	• Non-inert C&D materials ➢ Top soil, vegetation and wood waste, etc. ➢ Bamboo, timber, paper and plastic, etc.	• 238,600 m^3	• Reusable materials should be separated and recycled as far as practicable	• Reused on-site as much as possible. Materials that cannot be reused nor recycled will be disposed of at the WENT Landfill
		• Inert C&D materials ➢ Soft materials ➢ Artificial hard materials ➢ All grade granite	• 954,200 m^3	• Reusable materials should be separated and recycled as far as practicable	• Sorted materials will be used for construction by other concurrent projects or disposed of at Tuen Mun Area 38 Fill Bank for beneficial use
Chemical Waste	• Building demolition • Plant operation and maintenance • Maintenance of mechanical equipment	• Asbestos containing materials • Oil and grease, scrap batteries, used paint, fuel, etc. • Cleansing fluids and solvents from construction plant and equipment	• A few cubic metres per month	• Stored in compatible containers in designated area on-site • Collected by licensed collectors	• Recycled by licensed facility and/or disposal of at the CWTC

Waste Type	Generated from	Materials Generated	Total Amount Generated	Handling Procedures	Disposal Routes
General Refuse	• Refuse generated from construction works and site-based staff and workers.	• Food waste, containers, cans and waste paper, etc.	• Around 1,950 kg per day	• Provided on-site collection points together with recycling bins • Collected by a licensed collector	• Recycled at recycling facilities and/or disposed of at the NWNT RTS and/or the WENT Landfill
Excavated Sediment	• Excavated sediment generated from the inland water removal / diversion works.	• Land-based sediment	• Minimal	• All excavated sediment generated from the inland water removal / diversion works should be collected and handled in compliance with the Waste Disposal Ordinance	• The excavated sediment should be stabilised and solidified for reuse on site
Contaminated Soil	• Contaminated soil from existing land uses, e.g. vehicle maintenance workshop, open area storage, container storage and recycling facility, etc.	• Contaminated soil	• TBC	• Specific site investigation plans, including the proposed soil and groundwater sampling and testing strategy, were presented in the Contamination Assessment Plan (see **Appendix 8.1**)	• The contaminated soil should be remediated and recused onsite as far as practical; and landfills disposal should be adopted only as a last resort in accordance with Section 4.1 of the Practice Guideline for Investigation and Remediation of Contamination Land

Operation Phase

7.5.34 In accordance with Annex 7 and Annex 15 of the EIAO-TM as well as the requirements set out under Clause 3.4.8 of the EIA Study Brief (No. ESB-291/2015), an application for an EP will be submitted for a number of Schedule 2 DPs under this Project. The following activities, including the DPs identified under Schedule 2 and Schedule 3 and some non-DP elements in the Revised RODP (**Sections 1.5 and 2.4**), have been included in the waste management implication assessment for the operational phase:

- Construction of a new FWSR;
- Construction of a flood retention lake and flood retention facilities;
- Construction of a new DCS – subject to further review;
- Construction of a new primary distributor road (Road P1) (DP1);
- Construction of eight new distributor roads (Roads D1 to D8) (DP2);
- Construction of a new West Rail HSK Station (Site 4-34) (DP3);
- Construction of EFTS (DP4) – subject to further review;
- Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and KSWH connection to Road D3 (DP5);
- Construction of partly depressed and partly decked-over roads located at Road D2, Road D4 and Road D6 (DP6);
- Construction of a new container back-up and storage area (Sites 3-1, 3-4, 3-5, 3-13 and 3-14) (DP7) – subject to further review;
- Construction of a new HSK STW (Site 3-26 and part of existing San Wai STW) (DP8);
- Construction of four new SPSs (Sites 2-34, 3-41, 3-48 and 4-35) (DP9);
- Construction of Flushing Water Service Reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40) (DP10);
- Construction of a new RTS (Site 3-12) (DP11);
- Construction of Road P1 and a slip-road from KSWH to Road D3 partly located within the "CA" of Yuen Tau Shan (DP12);
- Other concurrent construction works; and
- "R" development, "G/IC", "E", "GB", open space, "A", other specified uses, industry, roads and "V".

7.5.35 The operation phase activities to be carried out for the Project will generate a variety of waste types. Typical waste types arising from the proposed works are identified in this section, together with an evaluation of the potential waste management impacts associated with the handling and disposal of waste. **Table 7.9** listed out the sources and examples of the identified waste types.

Table 7.9 Identification of Waste Types during the Operation Phase of the Project

Waste Type	Source of Waste	Example of Waste
Municipal Solid Waste (MSW)	• Domestic waste generated from future residences of public and private housing • Commercial and industrial (C&I) waste generated from enterprise and technology parks, offices and recreation sites	• Food waste, containers, cans and waste paper, etc. • Scrap materials, e.g. metals, etc.
Chemical Waste	• Chemical waste generated from ➢ Public facilities operation (e.g. STW, liquefied petroleum gas filling stations, petrol filling stations, etc.) ➢ Maintenance activities (e.g. buildings, infrastructure, roads, etc.)	• Paint, lubricants and used batteries, etc.
Screenings, Grits and Sewage Sludge	• Screenings and grits generated from sewage treatment process • Dewatered sludge generated from sewage treatment process	• Screenings and grits • Dewatered sludge

Municipal Solid Waste

7.5.36 The generation of MSW will be of most concern during the operation stage, which comprise domestic waste and C&I waste. **Table 7.10** shows the historical geographical variation in waste arising between 2008 and 2012.

Table 7.10 Historical Geographical Variation in Waste Arising (2008 – 2012)

	Average Domestic Waste (kg/person/day)					Average Commercial & Industrial Waste (kg/employee/day)				
District	**2008**	**2009**	**2010**	**2011**	**2012**	**2008**	**2009**	**2010**	**2011**	**2012**
Yuen Long	0.97	0.91	0.99	0.95	1.01	0.65	0.82	0.92	1.02	1.07
New Territories	0.78	0.81	0.79	0.76	0.79	0.74	0.77	0.78	0.79	0.78
Hong Kong	0.88	0.87	0.88	0.84	0.89	0.81	0.80	0.82	0.81	0.79

Sources:
1. Monitoring of Solid Waste in Hong Kong 2008 – 2012 (EPD, 2009 – 2013).
2. The Profile of Hong Kong Population analysed by District Council District, 2008 – 2010, 2012 (CSD, 2009 – 2011, 2013).
3. Population Census Summary Results, 2011 (CSD, 2012).

7.5.37 For a worst case assumption on future MSW generation, the per capita generation rates of domestic and C&I waste during the development period (2024 – 2038) have been calculated based on the average annual change of MSW generation at Yuen Long District using linear regression model. This model is applied uniformly to the year of completion for each development stage, without regard to variation within socio-economic groups.

Domestic waste $Yt = 0.0011(t – 7) + 0.9311$

C&I waste $Yt = 0.1042(t – 7) + 0.5813$

Where: Yt = waste generation rate at year t (in kg/person/day for domestic waste and in kg/employee/day for C&I waste)

t = year in last two figures of the year (e.g. “08” in 2008).

7.5.38 **Table 7.11** shows the calculated per capita domestic and C&I waste generation rates for each development stage (2024 – 2038) based on the average annual change of MSW generation at Yuen Long District using linear regression model.

Table 7.11 Calculated Per Capita Domestic and C&I Waste Generation Rates for Each Development Stage (2024 – 2038)

Development Stage (Year of Population Intake)	Domestic Waste Generation Rates (kg/person/day)	C&I Waste Generation Rates (kg/employee/day)
Advance Works – Phase 1 & 2 (2024)	1.12	2.35
Advance Works – Phase 3 (2030)	1.18	2.98
Stage 1 (2025)	1.13	2.46
Stage 2 (2031)	1.20	3.08
Stage 3 (2035)	1.24	3.50
Stage 4 (2038)	1.27	3.81

7.5.39 The Project targets to accommodate an additional of approximately 176,000 residents and 150,000 employees upon development completion. Based on the calculated domestic and C&I waste generation rates for each development stage as shown in **Table 7.11**, the estimated domestic and of waste generated in the development stages are provided in **Table 7.12**.

Table 7.12 Projected Quantities of Domestic and C&I Waste Arising for Each Development Stage (2024 – 2038)

Development Stage (Year of Population Intake)	Strategic Residents (Population)	Projected Domestic Waste (tonnes/day)	Strategic Employees (Population)	Projected C&I Waste (tonnes/day)	Total Projected Municipal Solid Waste (tonnes/day)
Advance Works – Phase 1 & 2 (2024)	6,551	7	322	1	8
Advance Works – Phase 3 (2030)	35,591	42	15,016	45	87
Stage 1 (2025)	150	0	4,477	11	11
Stage 2 (2031)	52,800	63	65,968	203	266
Stage 3 (2035)	44,225	55	58,387	204	259
Stage 4 (2038)	36,829	47	5,382	21	68
Total	**176,146**	**214**	**149,552**	**485**	**699**

7.5.40 As shown in **Table 7.12**, the projected domestic waste requiring disposal is 214 tonnes per day and the projected C&I waste requiring disposal is 485 tonnes per day. As such, the total MSW requiring disposal will be 699 tonnes per day.

7.5.41 An effective and efficient waste handling system is essential in order to minimise potential adverse environmental impacts during waste storage, collection and transport. Such impacts may include odour if waste is not collected frequently; water quality if waste enters storm water drains; aesthetics and vermin problems if the waste storage area is not well maintained and cleaned regularly. The waste handling system may also facilitate materials recovery and recycling.

7.5.42 A refuse collection room should be installed in each building at the ground floor for localised refuse collection. Also, options for adoption of automatic refuse collection system and organic waste treatment facilities within the residential and commercial complexes will be considered.

7.5.43 In accordance with Chapter 9 of the Hong Kong Planning Standards and Guidelines, a refuse collection point (RCP) is required to serve the needs of each population of 20,000 persons or areas within a distance of 500 metres, whereas a RTS is required to provide a handling capacity between 100 and 1,000 tonnes in New Town Areas, which is equivalent to between 100,000 and 1,000,000 population. To cope with the new population waste generation and to support the existing NWNT RTS, Sites 1-1, 3-46 and 5-29 have been allocated for construction of three new RCPs and Site 3-12 has been allocated for construction of a new RTS. **Figure 7.1** shows the location plan of the new RCPs and RTS.

7.5.44 Based on the latest construction programme for the development of the new RCPs and RTSs as shown in the Revised RODP, the two new RCPs in Sites 3-46 and 5-29 are scheduled to commission under Advance Works Phase 3 and an addition RCP in Site 1-1 is scheduled to commission under Stage 3 Works. On the other hand, the new RTS in Site 3-12 is schedule to commission under Stage 2 Works. Waste recycling facilities / containers are recommended to be included in the RCPs and RTS. A Community Green Station is proposed to be co-located with the new RTS for environmental education purpose and convenient collection of recyclables from the local community and to provide synergy to achieve better operational efficiency and environmental sustainability. It is expected that there will be sufficient refuse transfer capacity to handle the general refuse arising during the operation phase of the Project. The MSW will be transported to the RCPs or RTS then to the WENT Landfill for disposal.

7.5.45 The waste should be sorted to recover materials (such as paper, aluminium cans, plastic bottles and glass bottles, etc.) as far as possible before disposal at the landfill. Different containers should be provided for the storage of different recyclable materials (e.g. fluorescent lamps, toner cartridges, rechargeable batteries, scrap electrical and electronic appliances, etc.). To avoid potential odour nuisance to the residents during transport of waste, enclosed waste collection trucks should be used and the collection route and time should be properly planned. The new RCPs and RTS should contain compactors and/or related equipment to provide adequate waste handling services within the HSK NDA and odour treatment units to process the air before discharging to the environment. At least daily collection should be arranged by the waste collectors.

7.5.46 Mitigation and control requirements for MSW are detailed in **Sections 7.6.18 to 7.6.19**. Provided that the handling, storage and disposal of MSW are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of MSW during the operation phase of the Project are not expected.

Chemical Waste

7.5.47 Chemical waste such as paints, lubricants and used batteries may be generated during maintenance activities. This waste may pose environmental, health and safety hazards. Measures as stipulated in the Waste Disposal (Chemical Waste) (General) Regulation and the Code of Practice on the Packaging, Labelling and Storage of Chemical Waste should be strictly followed for the handling and disposal of chemical waste. The quantity of chemical waste to be generated during the operation is expected to be small and in the order of a few cubic metres per month.

7.5.48 Should any chemical waste be generated, the operator must register with EPD as a chemical waste producer. The chemical waste would be readily accepted for disposal of at the CWTC in Tsing Yi. This chemical waste should be collected periodically in drum-type containers by licensed chemical waste collectors. Mitigation and control requirements for chemical waste are detailed in **Section 7.6.20**. Provided that the handling, storage and disposal of chemical waste are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of chemical waste during the operation phase of the Project are not expected.

Screenings, Grits and Sewage Sludge

7.5.49 The proposed HSK STW is designed to handle the sewage arising from the Project. Tertiary treatment and secondary plus treatment are assumed for the proposed HSK STW subject to detailed design. The design capacity of HSK STW is 85,000 m^3 per day and the major solid waste types produced from the operation are screenings and grits collected from the inlet works and dewatered sludge associated with the sewage treatment process. It is estimated that around 16 m^3 per day of screenings and grits would be generated from the proposed new HSK STW during the operation phase of the Project. Screenings and grits should be properly stored in a covered container and disposed of daily to the WENT Landfill. The transportation and disposal of the screenings and grits will be managed and controlled by a reputable waste collector employed by the operator to reduce any potential pest odour and litter impacts.

7.5.50 Sludge thickening tank is suggested to reduce the volume of primary sludge generated from primary sedimentation tank in new HSK STW. Anaerobic digester is suggested to further reduce the sludge volume by about 50%. The biogas generated in the anaerobic digestion would be collected and reused in the anaerobic digester and for site use if sufficient. Centrifuge or filter press is suggested for digested sludge dewatering up to 30% w/w dry solid before disposal off-site. It is estimated that around 50 m^3 per day of dewatered sludge would be generated from the proposed new HSK STW during the operation phase of the Project. Dewatered sludge should be properly stored in a covered container after the sludge dewatering process and disposed of daily to the STF in Tuen Mun. The dewatered sludge will be delivered by road transport in water tight containers or skips to avoid odour emission during transportation to the STF. Mitigation and control requirements for screenings, grits and sewage sludge are detailed in **Section 7.6.21**. Provided that the handling, storage and disposal of screenings, grits and sewage sludge are in accordance with these requirements, the potential waste management implications, including potential hazards, air and odour emissions, noise and wastewater discharge, associated with handling, storage and disposal of screenings, grits and sewage sludge during the operation phase of the Project are not expected.

Transportation Arrangement for Waste Disposal During Operation Phase

7.5.51 Land transport should be used to deliver and dispose of the waste generated from the Project area to the designated disposal outlets. It is expected there will be a maximum of 50 vehicles per day for transporting waste during the operation phase of the Project. The transportation routings for the disposal of various types of waste generated during the operation phase of the Project are shown in **Table 7.13**. The transportation routings may change subject to the actual traffic conditions of the roads. Nevertheless, with the implementation of appropriate mitigation measures (e.g. using water-tight containers and covered trucks), no adverse environmental impacts are expected due to the transportation of waste.

Table 7.13 Tentative Transportation Routings for Waste Disposal During Operation Phase

Disposal Outlet	Type of Waste	Tentative Transportation Routing
WENT Landfill	MSW, Screenings and Grits	Via Kong Sham Western Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road and Nim Wan Road
CWTC	Chemical Waste	Via Kong Sham Western Highway, Yuen Long Highway, Tuen Mun Road, Tsing Long Highway, Tsing Sha Highway and Tsing Yi Road
STF	Dewatered Sludge	Via Kong Sham Western Highway, Yuen Long Highway, Tuen Mun Road, Wong Chu Road, Lung Fu Road, Lung Mun Road, Lung Kwu Tan Road and Nim Wan Road

Operation Phase Waste Summary

7.5.52 **Table 7.14** provides a summary of the waste types likely to be generated during the construction phase of the Project, together with the recommended handling and disposal methods.

Table 7.14 Summary of Waste Arising, Waste Handling Procedures and Disposal Routes during the Operation Phase of the Project

Waste Type	Generated from	Materials Generated	Total Amount Generated	Handling Procedures	Disposal Routes
Municipal Solid Waste (MSW)	• Domestic waste generated from future residences of public and private housing • Commercial and industrial (C&I) waste generated from enterprise and technology parks, offices and recreation sites	• Food waste, containers, cans and waste paper, etc. • Scrap materials, e.g. metals, etc.	• Domestic and C&I waste: Around 700 tonnes per day	• Provided on-site collection points together with recycling bins • Collected by a licensed collector	• Recycled at recycling facilities and/or disposed of at the new RCPs, RTS and/or the WENT Landfill
Chemical Waste	• Chemical waste generated from ➢ Public facilities operation (e.g. STW, liquefied petroleum gas filling stations, petrol filling stations, etc.) ➢ Maintenance activities (e.g. buildings, infrastructures, roads, etc.)	• Paint, lubricants and used batteries, etc.	• A few cubic metres per month	• Stored in compatible containers in designated area on-site • Collected by licensed collectors	• Recycled by licensed facility or disposed of at the CWTC
Screenings, Grits and Sewage Sludge	• Screenings, grits and dewatered sludge generated from STW	• Screenings and grits • Dewatered sludge	• Screening and grits: Around 16 m^3 per day • Dewatered sludge: Around 50 m^3 per day	• Stored in bins or other containers in designated area on-site • Collected by licensed collectors	• Screenings and grits are disposed of at the WENT Landfill • Dewatered sludge are disposed of at the STF

7.6 Mitigation of Adverse Waste Management Implications

General

Waste Management Hierarchy

7.6.1 The waste management hierarchy has been applied in the assessment and development of mitigation measures for waste which aims at evaluating the desirability of waste management methods and includes the following in descending preference:

- Avoidance and minimisation of waste generation;
- Reuse of materials as far as practicable;
- Recovery and recycling of residual materials where possible; and
- Treatment and disposal of waste according to relevant laws, guidelines and good practices.

7.6.2 Recommendations of good site practices and waste reduction measures should be stated in order to achieve avoidance and minimisation of waste generation in the waste management hierarchy. An EMP and trip-ticket system are recommended for monitoring management of waste. Specific measures targeting the mitigation of impacts in works areas and the transportation of waste off-site should be provided to minimise the potential impacts to the surrounding environment.

Good Site Practices

7.6.3 Good site practices will be included as part of the contract requirements to be addressed during the detailed design stage of the development by the Contractor. Adverse waste management implications are not expected provided that good site practices are strictly implemented. The following good site practices are recommended during the construction phase:

- Nomination of an approved personnel, such as a site manager, to be responsible for the implementation of good site practices;
- Training of site personnel in site cleanliness, proper waste management and chemical handling procedures;
- Provision of sufficient waste disposal points and regular collection of waste for disposal;
- Adoption of appropriate measures to minimise windblown litter and dust during handling, transportation and disposal of waste; and
- Preparation of a WMP in accordance with the ETWB TCW No. 19/2005 Environmental Management on Construction Sites and submitted it to the Engineer for approval.

Waste Reduction Measures

7.6.4 Amount of waste generation can be significantly reduced through good management and control. Waste reduction is best achieved by proper planning and design at the planning and design phases, as well as by ensuring the implementation of good site practices. The following recommendations are proposed to achieve waste reduction:

- Segregate and store different types of waste in different containers, skips or stockpiles to enhance reuse or recycling of materials and their proper disposal;

- Adopt proper storage and site practices to minimise the potential for damage to, and contamination of, construction materials;
- Plan the delivery and stock of construction materials carefully to minimise the amount of waste generated;
- Sort out demolition debris and excavated materials from demolition works to recover reusable / recyclable portions (i.e. soil, rock, broken concrete, etc.);
- Maximise the use of reusable steel formwork to reduce the amount of C&D materials;
- Minimise over ordering of concrete, mortars and cement grout by doing careful check before ordering; and
- Adopt pre-cast construction method instead of cast-in-situ method for construction of concrete structures as far as possible.

Storage, Collection and Transportation of Waste

7.6.5 Storage of materials on-site may induce adverse environmental impacts if not properly managed. The following recommendations should be implemented to minimise the impacts:

- Waste, such as soil, should be handled and stored well to ensure secure containment, thus minimising the potential of pollution;
- Maintain and clean storage areas routinely;
- Stockpiling area should be provided with covers and water spraying system to prevent materials from being wind-blown or washed away; and
- Different locations should be designated to stockpile each material to enhance reuse.

7.6.6 Waste hauler with appropriate permits should be employed by the Contractor for the collection and transportation of waste from works areas to respective disposal outlets. The following recommendation should be implemented to minimise the impacts:

- Remove waste in timely manner;
- Employ the trucks with cover or enclosed containers for waste transportation;
- Obtain relevant waste disposal permits from the appropriate authorities; and
- Dispose of waste at licensed waste disposal facilities.

Construction Phase

Construction and Demolition Materials

7.6.7 Wherever practicable, C&D materials should be segregated from other waste to avoid contamination and ensure acceptability at the public filling areas or reclamation sites. The following mitigation measures should be implemented in handling the C&D materials:

- Adopt “selective demolition” technique to demolish the existing structure and facilities with a view to recovering broken concrete effectively for recycling purpose, where possible;

- Maintain the stockpile areas and reuse excavated fill material for backfilling;
- Carry out on-site sorting to recover the inert C&D materials and reusable and recyclable materials prior to disposal off-site;
- Make provisions in the contract documents to allow and promote the use of recycled aggregates where appropriate; and
- Implement a trip-ticket system for each works contract in accordance with DEVB TC(W) No. 6/2010 Trip-ticket System for Disposal of Construction and Demolition Material to ensure that the disposal of C&D materials are properly documented and verified.

7.6.8 The Contractor should be responsible for devising a system to work for on-site sorting of C&D materials. It is recommended that the system should include the identification of the source of generation, estimated quantity of waste generated, arrangement for on-site sorting and/or collection, designated stockpiling areas, frequency of collection by recycling contractors and frequency of removal off-site.

Chemical Waste

7.6.9 Due to the potential large amount of ACM during the site clearance stage, asbestos investigation is required. However, as asbestos investigation will involve a large number of buildings and most premises will involve private access, which cannot be obtained at this stage, it is considered that an asbestos specialist shall be employed by the responsible parties during the construction stage to investigate this issue.

7.6.10 Sufficient and reasonable lead time shall be allowed for preparation, vetting and implementation of Asbestos Investigation Report and Asbestos Abatement Plan in accordance with Air Pollution Control Ordinance before commencement of any demolition or site clearance work.

7.6.11 Some key precautionary measures related to the handling and disposal of asbestos are listed as following:

- Adoption of protection, such as full containment, mini containment, or segregation of work area;
- Provision of decontamination facilities for cleaning of workings, equipment and bagged waste before leaving the work area;
- Adoption of engineering control techniques to prevent fibre release from work area, such as use of negative pressure equipment with high efficiency particulate air (HEPA) filters to control air flow between the work area and the outside environment;
- Wetting of asbestos containing materials before and during disturbance, minimising the breakage and dropping of asbestos containing materials, and packing of debris and waste immediately after it is produced;
- Cleaning of work area by wet wiping and vacuuming with HEPA-filtered vacuum cleaner;
- Coating on any surfaces previously in contact with or contained by asbestos with a sealant;
- Proper bagging, safe storage and disposal of asbestos and asbestos-contaminated waste;
- Pre-treatment of all effluent from the work area before discharged; and

- Air monitoring strategy to check the leakage and clearance of the work area during and after the asbestos work.

7.6.12 The handling and disposal of ACM will be carried out in accordance with the EPD's Code of Practice on Handling, Transportation and Disposal of Asbestos Waste and ProPECC PN 2/97 Handling of Asbestos Containing Materials in Buildings.

7.6.13 For those processes which generated chemical waste, it may be possible to find alternatives to eliminate the use of chemicals, to reduce the generation quantities or to select a chemical type of less impact on environment, health and safety as far as possible.

7.6.14 If chemical waste is produced at the construction site, the Contractor will be required to register with the EPD as a chemical waste producer and to follow the guidelines stated in the Code of Practice on the Packaging, Labelling and Storage of Chemical Waste. Chemical waste should be stored in appropriate containers and collected by a licensed chemical waste contractor. Chemical waste (e.g. spent lubricant oil) should be recycled at an appropriate facility as far as possible, while chemical waste that cannot be recycled should be disposed of at either the CWTC, or another licensed facility, in accordance with the Waste Disposal (Chemical Waste) (General) Regulation.

General Refuse

7.6.15 General refuse should be stored in enclosed bins separately from construction and chemical waste. Recycling bins should also be placed to encourage recycling. Preferably enclosed and covered areas should be provided for general refuse collection and routine cleaning for these areas should also be implemented to keep areas clean. A reputable waste collector should be employed to remove general refuse on a daily basis. It is expected that such arrangements would minimise potential environmental impacts.

Excavated Sediment

7.6.16 Since the amount of excavated sediment generated from the inland water removal / diversion works is expected to be small, all excavated sediment will be treated and reused on-site as backfilling materials for the Project. This approach avoids the need for off-site disposal that may result in impacts on the marine environment. In addition, all construction works near the watercourses should be undertaken within a dry zone and during dry season to avoid adverse impacts to the environment. The excavated sediment, if stockpiled on site, should be stored in enclosed containers and transported to the on-site treatment facilities as soon as practicable to minimise any potential odour impacts.

Contaminated Soil

7.6.17 It is considered unlikely that contaminated land issues, if any subject to site investigation, would be a concern during either the construction or the operational of the proposed development as remediation on contaminated area would be carried out prior to construction. However, as a precaution, it is recommended that standard good site practices should be implemented during the construction phase to minimise any potential exposure to contaminated soils or groundwater.

Operation Phase

Municipal Solid Waste

7.6.18 Implementation of a waste prevention programme as well as materials recovery and recycling programme are recommended in order to minimise the production of waste. The programmes should consist of the following components:

- Recycling bins such as paper, aluminium cans, plastic bottles, glass bottles, etc. should be placed at prominent locations to encourage recycling;
- Banner should be erected at the recycling bins area;
- Operator should make arrangements with the recycler to collect and recycle used fluorescent lamps, toner cartridges as well as the scrap electronic equipment, such as computers to avoid disposal at landfills as far as practicable;
- Staff awareness training should be provided on waste management procedures, including waste reduction and recycling;
- Operator should set up waste reduction and recycled targets; and
- Operator should participate in the Wastewi$e Label scheme to facilitate waste reduction.

7.6.19 MSW generated from residential, commercial and industrial buildings should be collected with lidded bins, delivered to the refuse collection room and stored in enclosed containers installed in each building at the ground floor to prevent windblown, vermin, water pollution and visual impact. At least daily collection should be arranged by the waste collector to transport the waste to the RCPs or RTS within the HSK NDA. Odour removal installations are recommended to be installed at the RCPs and RTS to treat the exhaust air. Wastewater generated at the RTS should be treated at the on-site wastewater treatment plant prior to discharge to the public sewerage systems. Such arrangements will minimise potential environmental impacts. Furthermore, the low emission truck, such as EURO V or later model is recommended to be used for waste transportation to minimise traffic emission and the potential air quality impacts. The above recommendations are proposed as technical guidelines for the operator's consideration and will be subject to detailed design.

Chemical Waste

7.6.20 The proposed mitigation measures for operation phase are the same as that proposed for the construction phase. The operator should register with EPD as a chemical waste producer and follow the guidelines stated in the Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes. Chemical waste should be stored in appropriate containers and collected by a licensed chemical waste contractor. Chemical waste (e.g. spent lubricant oil) should be recycled at an appropriate facility as far as possible, while chemical waste that cannot be recycled should be disposed of at either the CWTC, or another licensed facility, in accordance with the Waste Disposal (Chemical Waste) (General) Regulation.

Screenings, Grits and Sewage Sludge

7.6.21 The new HSK STW is designed to handle the sewage generated from the new development areas under this Project. The major solid waste types produced from the STW would be the screenings and grits collected from the inlet works and the dewatered sludge collected from the sewage treatment process. Screenings and grits generated from the STW is suggested to be disposed of at the WENT Landfill whereas the dewatered sludge generated from the STW is suggested to be treated at the STF. The screenings, grits and dewatered sludge will be delivered by road transport in water tight containers or skips to avoid odour emission during transportation. Unloading process will be operated in the designated room inside STW which should be enclosed and served by negative pressure by extracting odorous gas to deodorising unit.

7.7 Evaluation of Residual Environmental Impacts

7.7.1 With the implementation of recommended mitigation measures for the handling, transportation and disposal of the identified waste, adverse residual waste management implications are not anticipated during both the construction and operational phases.

7.8 Environmental Acceptability of the Schedule 2 Designated Projects

7.8.1 Waste management implication assessment for the construction and operation phases of the Project was carried out taking into account the DPs under Schedule 2 and Schedule 3 (see **Sections 7.5.1 and 7.5.36**). It is noted that the DPs under Schedule 2 would mostly contributed to the generation of C&D materials during the construction phase of the Project. The majority of the non-inert C&D materials will be generated from demolition of existing structures and tree felling, while the majority of the inert C&D materials will be generated from refinement of site formation level and the proposed construction activities. Since most of the inert C&D materials generated from the Project will be reused as backfilling materials, the DPs under Schedule 2 are not anticipated to result in any adverse environmental impacts.

7.9 Cumulative Impacts from Concurrent Project

7.9.1 No major concurrent project related to waste management implication assessment is identified within the study area; therefore, no adverse cumulative waste management implications would be expected.

7.10 Conclusion

7.10.1 The main waste types to be generated during the construction phase of the Project will include C&D materials, chemical waste, general refuse, excavated sediment and contaminated soil. Reduction measures have been recommended to minimise the amount of materials generated by the Project by reusing C&D materials as far as practicable before off-site disposal. Provided that the waste is handled, transported and disposed of using approved methods, adverse environmental impacts would not be expected.

7.10.2 The main waste types to be generated during the operation phase of the Project will include MSW, chemical waste, screenings, grits and sewage sludge. Three new RCPs and a new RTS have been included in the Revised RODP in preparation for the increased quantity of waste in the district. A Community Green Station is proposed to be co-located with the new RTS for environmental education purposes, convenient collection of recyclables from the local community, and to provide synergy to achieve better operational efficiency and environmental sustainability. Provided that the waste is handled, transported and disposed of using approved methods, adverse environmental impacts would not be expected.